

CardioIA — Technical Report

Phase 7: Full Cardiac Intelligence Platform 2TIAOR — FIAP 2026

1. Introduction

CardioIA Phase 7 consolidates all previous phases into a unified Full Cardiac Intelligence Platform. The project transitions from isolated AI modules to an integrated digital product: a web dashboard, mobile app, Python backend, and IoT sensor layer — all connected through a single REST API.

2. Architecture Overview

The platform follows a hub-and-spoke architecture with the **FastAPI backend** as the central hub:

- **IoT Layer** (MicroPython on ESP32/Wokwi) → sends sensor data to backend
- **Backend** (FastAPI on Render) → orchestrates AI engines, serves REST API
- **Web** (React + Vite + shadcn/ui on Vercel) → consumes API, renders dashboard
- **Mobile** (React Native + Expo) → same API consumption, native Android experience

See `docs/architecture.md` for the full Mermaid diagram.

3. Backend Integrator

The FastAPI backend exposes 7 endpoint groups:

Endpoint	Purpose
GET /health	Service health check
POST /api/auth/login	Mock authentication (admin/cardioai)
POST /api/predict	Cardiac risk prediction using Phase 6 RF model
POST /api/chat	LLM-powered cardiac assistant via OpenRouter
POST /api/sensors	IoT sensor data ingestion
GET /api/sensors/latest	Retrieve latest sensor readings
GET /api/symptoms/*	Knowledge base search (Phase 2 data)

Key design decisions: - OpenRouter API key lives exclusively on the backend (Render env vars) - The RF model (`cardio_risk.joblib`) is loaded once at startup - Sensor data is stored in-memory (deque of 100 readings) for MVP simplicity - CORS is configured for both local dev and production origins

4. IoT Module (MicroPython)

The Phase 3 Arduino C++ sketch (1475 lines) was converted to MicroPython (~290 lines) for the ESP32, simulated on Wokwi:

- **DHT22** — temperature sensor
- **Potentiometer** — simulates heart rate (mapped to 40-180 BPM)
- **SSD1306 OLED** — displays real-time readings
- **3 LEDs** (Green/Yellow/Red) — visual alert based on thresholds
- **HTTP POST** — sends readings to backend every 3 seconds

Thresholds trigger alerts: high temperature ($>38^{\circ}\text{C}$), abnormal heart rate (<50 or >120 BPM).

5. Web Dashboard

Built with React 18 + Vite + TypeScript + Tailwind CSS v4 + shadcn/ui:

- **Login** — mock authentication with session persistence
- **Dashboard** — sensor monitoring cards (temperature, heart rate, IoT alerts), risk prediction form with 6 clinical inputs, protocol recommendations
- **Chat** — conversational AI assistant with message history

Deployed on Vercel with `vercel.json` configured for SPA route rewrites.

6. Mobile App

Built with React Native + Expo SDK 56 + React Native Paper (Material Design 3):

- Same feature set as web: Login → Dashboard → Chat
- Environment configured via `app.config.js` + `.env` (`EXPO_PUBLIC_API_URL`)
- APK generated via EAS Build (`eas.json` preview profile)
- Android package: `br.com.fiap.cardioai`

7. Cross-Phase Integration

Phase	Artifact	Integration Point
Phase 2	<code>knowledge_map.csv</code>	Chat context enrichment + symptom search API
Phase 3	<code>sketch.ino</code>	Converted to MicroPython <code>main.py</code>
Phase 6	<code>cardio_risk.joblib</code>	<code>/api/predict</code> endpoint
Phase 6	<code>protocols.json</code>	Risk-based clinical recommendations

8. Deployment

Component	Platform	URL
Backend	Render (free tier)	https://twotiaor-fase7-cap1.onrender.com
Web	Vercel	https://2-tiaor-fase7-cap1.vercel.app/
Mobile	Expo EAS Build	https://expo.dev/accounts/gabem/5e6a-43a5-8606-1ded7d704d28
IoT	Wokwi	https://wokwi.com/projects/4662

Video Demo: <https://youtu.be/USCApx9ZqKM>

9. Conclusion

Phase 7 demonstrates the full lifecycle of an AI-powered healthcare platform: from edge sensor data collection (IoT/MicroPython), through backend orchestration (FastAPI/Python), to user-facing interfaces (React web + React Native mobile). The architecture prioritizes security (API keys backend-only), simplicity (single REST API for all clients), and cross-phase reuse of previously developed AI models and knowledge bases.